

Motion Control #1: Joint and Operational Space

Optimization and Learning for Robot Control

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1. Joint Motion Control

Objective: Given a reference joint trajectory $q^{ref}(t)$, compute $\tau(t)$ such that $q(t) \simeq q^{ref}(t)$.

- **System Dynamics Assumption:** $M(q)\ddot{q} + h(q, \dot{q}) = \tau$.

- **EPID Control Law:**

$$\tau(t) = K_p e(t) + K_d \dot{e}(t) + K_i \int_0^t e(s) ds$$

where the error is defined as $e(t) \triangleq q^{ref}(t) - q(t)$.

- K_p, K_d, K_i are positive-definite diagonal gain matrices.

2. Inverse Dynamics Control

The goal is to achieve **Feedback Linearization** by compensating for non-linear terms.

Control Law:

$$\tau = M\ddot{q}^d + h$$

$$\ddot{q}^d = \ddot{q}^{ref} + K_d\dot{e} + K_p e$$

Closed-loop Error Dynamics: Substituting τ into the plant dynamics $M\ddot{q} + h = \tau$:

$$M\ddot{q} + h = M(\ddot{q}^{ref} + K_d\dot{e} + K_p e) + h$$

$$\implies \ddot{e} + K_d\dot{e} + K_p e = 0$$

The system behaves as a decoupled linear system.

3. Mathematical Insight: Pseudo-inverse ($J^\dagger$)

When a robot is **redundant** ($n > m$), the Jacobian $J \in \mathbb{R}^{m \times n}$ has no unique inverse.

Connection to Least Squares Method (LSM): Minimize the joint velocity norm while satisfying the kinematic constraint:

$$\min_{\dot{q}} \frac{1}{2} \dot{q}^T \dot{q} \quad \text{subject to} \quad \dot{x} = J\dot{q}$$

Using Lagrange Multipliers ($L = \frac{1}{2} \dot{q}^T \dot{q} + \lambda^T (\dot{x} - J\dot{q})$):

$$\textcircled{1} \quad \frac{\partial L}{\partial \dot{q}} = \dot{q} - J^T \lambda = 0 \implies \dot{q} = J^T \lambda$$

$$\textcircled{2} \quad \dot{x} = J(J^T \lambda) \implies \lambda = (JJ^T)^{-1} \dot{x}$$

$$\textcircled{3} \quad \therefore \dot{q} = J^T (JJ^T)^{-1} \dot{x} = J^\dagger \dot{x}$$

4. Weighted LSM and Dynamically Consistent Inverse

If we minimize a weighted norm (e.g., Kinetic Energy) $\frac{1}{2}\dot{q}^T W \dot{q}$:

$$\dot{q}_{opt} = W^{-1} J^T (J W^{-1} J^T)^{-1} \dot{x}$$

Dynamic Consistency: Choosing the mass matrix M as the weight ($W = M$) yields:

$$\bar{J} = M^{-1} J^T (J M^{-1} J^T)^{-1} = M^{-1} J^T \Lambda$$

where Λ is the **Operational Space Inertia Matrix**. This ensures that joint torques do not produce internal accelerations in the task space.

5. Operational Space Dynamics Derivation

Starting from $\ddot{x} = J\ddot{q} + \dot{J}\dot{q}$ and $M\ddot{q} + h = \tau$:

- 1 Express joint acceleration: $\ddot{q} = M^{-1}(\tau - h)$.
- 2 Substitute into task acceleration: $\ddot{x} = JM^{-1}(\tau - h) + \dot{J}\dot{q}$.
- 3 Pre-multiply by $\Lambda = (JM^{-1}J^T)^{-1}$:

$$\Lambda\ddot{x} + \underbrace{\Lambda(JM^{-1}h - \dot{J}\dot{q})}_{\mu} = \underbrace{\Lambda JM^{-1}\tau}_{F_{task}}$$

If we assume $\tau = J^T F_\tau$, then $F_{task} = F_\tau$.

6. Decoupling and Control Law

Control Law: $\tau = J^T \Lambda \ddot{x}^d + h$.

Verification: Substitute τ into the system dynamics:

$$M\ddot{q} + h = J^T \Lambda \ddot{x}^d + h$$

$$J\ddot{q} = JM^{-1}J^T \Lambda \ddot{x}^d \implies J\ddot{q} = \ddot{x}^d$$

Since $\ddot{x} = J\ddot{q} + \dot{J}\dot{q}$, if $\dot{J}\dot{q} \simeq 0$, then $\ddot{x} \simeq \ddot{x}^d$.

Desired Task Acceleration: $\ddot{x}^d = \ddot{x}^{ref} + K_d(\dot{x}^{ref} - \dot{x}) + K_p(x^{ref} - x)$.

7. Redundancy Resolution

Using the **Dynamically Consistent Pseudo-inverse** $J^{T\#} = \Lambda JM^{-1}$:

$$\tau = J^T F_\tau + (I - J^T J^{T\#})\tau_1$$

The term $(I - J^T J^{T\#})$ is the **Null-space Projector**.

Proof of Decoupling:

$$J^{T\#}\tau_{null} = J^{T\#}(I - J^T J^{T\#})\tau_1 = (J^{T\#} - J^{T\#}J^T J^{T\#})\tau_1 = (J^{T\#} - J^{T\#})\tau_1 = 0$$

This proves that τ_1 does not affect the end-effector motion.

Secondary Task: $\tau_1 = M(K_p(q^{ref} - q) - K_d\dot{q}) + h$.